Inkloom AI — Brand-Governance Platform

Round 2 Build Report & Updated Technical Proposal

Date: 22–25 October 2026
Stage: Round 2 Final Submission

TEAM NAME	TRACK	TEAM MEMBERS	BUILD STATUS
indiedevs	Enterprise AI & Automation	Utkarsh Yadav, Shreya Kumari	Production Ready MVP

1. EXECUTIVE SUMMARY & PROGRESS FROM ROUND 1

In Round 1, team **indiedevs** proposed **Inkloom AI**: an intelligent brand-governance co-pilot that resolves the core friction of decentralized enterprise communications—outdated product claims, inconsistent tone of voice, non-inclusive terminology, visual design divergence, and accessibility failures.

For **Round 2 — The Build**, we built a **production-grade platform architected to scale to millions of users** with zero-cost cloud sustainability, sub-5ms deterministic performance, and zero-knowledge privacy guarantees. The product features full multi-format scanning (text, markdown, images), explainable rule citations, one-click fixes, role-based reviews, verifiable clearance certificates, real-time free-tier usage metering, and command palette navigation.

2. COMPLETED MVP FEATURES IN ROUND 2

1. Centralized Brand Rule Repository

Persistent storage with CRUD operations for Tone-of-Voice rules, verified Product Facts & SLAs, restricted vocabulary dictionaries (DEI & competitor rules), and approved visual palettes. Includes export/import and active rule toggles.

2. Multi-Format Real-Time Scanner

Dual-engine analysis for text copy, markdown files, and image assets. Evaluates formality, tone polarity, reading time, character offsets, WCAG 2.1 AA color contrast, and accessibility alt-text descriptive depth.

3. Explainable Flagging & Remediation

Pinpoints violations with exact line citations from the Brand Operations Manual, explains the legal/reputational brand risk, and offers context-preserving replacements with individual "Apply Fix" and "Fix All" actions.

4. Security & Zero-Knowledge Vault

Ephemeral Zero-Data Retention Mode (volatile in-memory scanning without disk writes), Automated Secret/PII Scrubber (AWS, API keys, emails), client-side WebCrypto SHA-256 signatures, and emergency data purge.

5. Sustainable Free Tier & Quota Meter

Live monthly usage tracker (10,000 words/mo included without credit card), interactive plan selector (Free Community, Pro Team $29/mo, Enterprise Scale), and zero-friction local-first processing.

6. Collaborative Review & Certification

Role-based access control (Creator Utkarsh, Brand Manager Shreya, Compliance Lead), document lifecycle tracking, visual diff mode, threaded comments, and signed Brand Clearance Certificates.

> **Hackathon Usability & Quick Access:** Built-in "Judge Demo Presets" selector (SaaS Launch, Security Whitepaper, Social Campaign, Governed Showcase) and `Cmd+K` Command Palette allow evaluators to test all edge cases and remediation pipelines in under 60 seconds.

3. COMPLETE END-TO-END PRODUCT WORKFLOW

1. **Brand Ingestion & Rule Setup:** Brand managers configure tone dimensions, verified SLAs (e.g. 99.95% standard), restricted terminology, and approved hex palettes in the Brand Rule Repository.
2. **Content Drafting & Upload:** Creators draft announcements or drop markdown/text files into the multi-channel editor (Marketing, Docs, Social, Press Release).
3. **Real-Time AI Governance Scan:** The system scans content simultaneously across Tone, Facts, Terminology, and Clarity, computing a weighted 0–100% Brand Health score with letter grades in <5ms.
4. **Explainable Remediation:** The copilot displays flagged snippets with rule citations, risk rationales, and one-click fixes. Users can trigger "Fix All" to automatically remediate all issues.
5. **Editorial Review & Formal Certification:** Drafts move through role approvals with change tracking and threaded discussions. Upon approval, a signed Brand Governance Clearance Certificate is issued.
6. **Macro Analytics & Continuous Improvement:** Aggregates department compliance trends, recurring violation categories, and the Guideline Clarity Index to identify training gaps.

4. TECHNOLOGY STACK & IMPLEMENTATION DETAILS

Layer	Technology	Implementation Purpose
Frontend UI/UX	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons	Executive enterprise dashboard with sub-50ms render latency, responsive dual-pane layout, dark-mode aesthetics, and Cmd+K command palette.
Backend API	Python 3.14, FastAPI, Uvicorn, Pydantic v2	High-concurrency asynchronous REST APIs for scanning, document workflow state transitions, usage metering, and guideline management.
AI Governance Engine	Hybrid Rule Heuristic + Semantic Claim Verifier + LLM Copilot	Deterministic regex & semantic boundary matching combined with optional Gemini/OpenAI API integration for context-aware explanations.
Security & Privacy	WebCrypto SubtleCrypto, Ephemeral RAM, PII Scrubber	Volatile in-memory execution for NDA drafts, automatic secret masking, and SHA-256 tamper-proof document signatures.
Visual & WCAG Engine	Pillow (PIL), Euclidean Color Space, WCAG 2.1 AA	Extracts dominant 5-color palette from graphics, compares to brand hex codes, verifies 4.5:1 contrast, and scores alt-text accessibility.
Data Persistence	SQLite / JSON-backed persistent store	Thread-safe persistence for rules, review drafts, revision diffs, threaded comments, and audit trails across restarts.

5. EXPLANATION OF CHANGES MADE TO ORIGINAL PLAN

- **Mobile-First Native App Experience & Smartphone Frame:** Redesigned the entire UI/UX to focus on mobile touch devices with a floating bottom glass navigation dock (framer-motion physics), thumb-zone action bars, segmented draft/flags/score tabs, and an interactive iPhone 16 Pro device frame toggle for desktop evaluators.
- **Added Sustainable Free Tier & Quota Meter:** Architected a zero-cost tier sustainable model (10,000 words/mo free without credit card) with an interactive tier selector and live quota gauge.
- **Added Security & Zero-Knowledge Vault:** Added an Ephemeral Mode toggle (zero disk persistence), automated secret/PII scrubber, and WebCrypto SHA-256 digital document signatures.
- **Added Multi-Modal Visual Asset Inspector in Round 2:** Accelerated from Round 3 to allow judges to evaluate visual brand consistency and WCAG 2.1 AA alt-text compliance immediately.
- **Added 1-Click "Fix All" Remediator & Command Palette:** Implemented reverse-offset multi-violation remediation and global `Cmd+K` keyboard navigation.

6. THREE MAJOR PLANNED IMPROVEMENTS FOR ROUND 3

1. **Deep Multi-Modal Visual Brand Engine (Logo Geometry, Clear Space & Vector Alignment):** Computer vision models that detect logo minimum clear-space violations, prohibited rotations, unapproved typography in marketing assets, and automated watermark placement.
2. **Omnichannel Workspace Integrations & Browser Extension:** Native plugins and a Chromium workspace extension allowing distributed teams to run Inkloom AI directly inside Google Docs, Notion, and Figma before copy is published.
3. **Autonomous Guideline Self-Optimization & Predictive Risk Engine:** An active learning feedback loop that analyzes reviewer override patterns and editorial disputes to automatically suggest style guide refinements to the Brand Governance Board.

7. USABILITY & JUDGING VERIFICATION GUIDE

Quick 60-Second Evaluation Path for Judges:

1. Open the application at `http://localhost:8000`.
2. Click **"Judge Demo Presets"** in the top header and select *"1. SaaS Launch Draft"*.
3. Observe the 42% compliance score, latency benchmark (<4ms), and 8 explainable violations with exact citations.
4. Click **"Fix All (8)"** in the top right to watch the AI remediator boost score to 98% with confetti celebration!
5. Click **"Submit for Review"**, then switch persona to *"Shreya Kumari (Brand Manager)"*.

6. Navigate to **"Review & Approvals"**, toggle **"View Changes Diff"**, and click **"Approve Content"**.
7. Switch to *"Compliance Officer"* and click **"View Brand Clearance Certificate"** to inspect the signed certificate!
8. Explore the **"Security & Vault"** tab to toggle Ephemeral Mode and view the threat model, or open **"Plans & Free Tier"** to inspect the live quota meter.